

Birthday attack- derivation

Module IV



The Birthday paradox

- Hash functions have collisions as a result of the basic hash function requirements. The first requirement specifies that $h(x)$ must accept inputs of arbitrary size.
- The second requirement specifies that $h(x)$ must generate an output of fixed size.
- Therefore, the size of the Input message space is significantly larger than the size of the output hashed value space.
- As a result, there must exist multiple messages that map to the same hashed value.
- The nature of collisions is illustrated with a phenomenon known as the Birthday Paradox.
- Consider a question, “how many people must be present so that there is a 50% probability that at least two of the people have the same birth date.” (Ignore leap year).
- There are 365 possible birth dates.

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- The probability that a person has a given birth date is $\frac{1}{365}$.
- Thus, the probability that no two people do not have the same birth date, i.e. that there is no collision is $1 - \frac{1}{365}$.
- In the case, where there are three people, the probability that no two people have the same birth date is:

$$p(\text{no collision}) = \left(1 - \frac{1}{365}\right) \cdot \left(1 - \frac{2}{365}\right)$$

- Similarly, when there are four people, the probability that no two people have the same birth date is:

$$p(\text{no collision}) = \left(1 - \frac{1}{365}\right) \cdot \left(1 - \frac{2}{365}\right) \cdot \left(1 - \frac{3}{365}\right)$$

- Finally, if there are p people, the probability that no two people have the same birth date is,

$$p(\text{no collision}) = \left(1 - \frac{1}{365}\right) \cdot \left(1 - \frac{2}{365}\right) \cdot \left(1 - \frac{3}{365}\right) \dots \left(1 - \frac{p-1}{365}\right)$$

$$p(\text{no collision}) = \prod_{i=1}^{p-1} \left(1 - \frac{i}{365}\right)$$

The Birthday paradox cntd..

- Finally, if there are p people, the probability that no two people have the same birth date is,

$$p(\text{no collision}) = \prod_{i=1}^{p-1} \left(1 - \frac{i}{365}\right)$$

Generalizing this equation for n people random values instead of 365 possible birth dates, the probability that no collision occurs if there are p random elements is ,

$$\prod_{i=1}^{p-1} \left(1 - \frac{i}{n}\right)$$

The equation further can be reduced by considering the Taylor Series:

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots$$

If $x \ll 1$, then $e^{-x} \approx 1 - x$.

Therefore, the probability that no collision occurs if there are p random elements may be rewritten as:

$$p(\text{no collision}) \approx \prod_{i=1}^{p-1} e^{-\frac{i}{n}} \approx e^{-\frac{1}{n}} \cdot e^{-\frac{2}{n}} \cdot e^{-\frac{3}{n}} \cdot \dots \cdot e^{-\frac{(p-1)}{n}} \approx e^{-\frac{(1+2+3+\dots+p-1)}{n}} \approx e^{-\frac{p \cdot (p-1)}{2n}}$$

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- Finally, if there are p people, the probability that no two people have the same birth date is,

$$p(\text{no collision}) \approx e^{-\frac{p \cdot (p-1)}{2n}}$$

Now, the probability of that at least one collision occurs determined as,

$$p(\text{at least one collision}) \approx 1 - p(\text{no collision})$$

$$p(\text{at least one collision}) \approx 1 - e^{-\frac{p \cdot (p-1)}{2n}}$$

If ε (*epsilon*), is used to denote the probability that at least one collision occurs, then:

$$\varepsilon \approx 1 - e^{-\frac{p \cdot (p-1)}{2n}}$$

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$$\rightarrow \varepsilon \approx 1 - e^{-\frac{p.(p-1)}{2n}}$$

$$\rightarrow e^{-\frac{p.(p-1)}{2n}} \approx 1 - \varepsilon$$

$$\rightarrow -\frac{p.(p-1)}{2n} \approx \ln(1 - \varepsilon)$$

$$\rightarrow p.(p-1) \approx -2n.\ln(1 - \varepsilon)$$

$$\rightarrow p^2 - p \approx 2n.\ln\left(\frac{1}{1-\varepsilon}\right)$$

For large values of p , i.e. a large number of random elements to be tested to find a collision, p^2 is significantly large than p .

Therefore,

$$p^2 \approx 2n.\ln\left(\frac{1}{1-\varepsilon}\right)$$

Therefore, the probability that at least one collision occurs is ,

$$p \approx \sqrt{2n.\ln\left(\frac{1}{1-\varepsilon}\right)}$$

Example1

- How many people must be present so that there is a 50% probability that at least two of the people have the same birth date if leap year ignored such that February 29th is not viable birth date, i.e. that $n=365$?

Solution: A 50% probability that at least two of the people have the same birth date results in $\varepsilon = 0.5$, i.e. that the probability that at least one collision occurs is 50%. Therefore, the number of people that must be present so that there is a 50% probability that at least two of the people have the same birth date is:

$$p \approx \sqrt{2n \cdot \ln\left(\frac{1}{1-\varepsilon}\right)} \rightarrow p \approx \sqrt{2 \cdot 365 \cdot \ln\left(\frac{1}{1-0.5}\right)} \rightarrow p \approx \sqrt{730 \cdot \ln\left(\frac{1}{0.5}\right)} \rightarrow p \approx \sqrt{730 \cdot \ln(2)}$$
$$p \approx \sqrt{730 \cdot 0.693} \rightarrow p \approx \sqrt{505.89} \rightarrow 22.5$$

Therefore, if there are approximately 23 people present, there is a 50% probability that at least two of the people have the same birth date.

Example2

The birthday paradox can be extended to Hash functions to determine how many random outputs must be generated such that the probability of a collision occurring is a specified percentage.

- How many random outputs are required so that there is a 50% probability of a collision occurring if an 80-bit Hash Function is used?

Solution: A 50% probability of a collision occurring results in $\varepsilon = 0.5$. Therefore, the number of random outputs that are required is:

$$p \approx \sqrt{2n \cdot \ln\left(\frac{1}{1-\varepsilon}\right)} \rightarrow p \approx \sqrt{2 \cdot 2^{80} \cdot \ln\left(\frac{1}{1-0.5}\right)} \rightarrow p \approx \sqrt{2 \cdot 2^{80} \cdot \ln(2)} \rightarrow p \approx \sqrt{2 \cdot 2^{80} \cdot 0.693}$$
$$p \approx \sqrt{2.2^{80} \cdot 0.693} \rightarrow p \approx \sqrt{2^{80} \cdot 1.386} \rightarrow 2^{40} \cdot \sqrt{1.386} \rightarrow 2^{40} \cdot 1.177$$

Therefore, there is a 50% probability of a collision occurring if there are approximately 2^{40} random Hash Function outputs for the given 80-bit hash function.

As a result of the Birthday Paradox, current Hash Functions have at least 256 output bits, resulting in the need For $p \approx 1.177 \cdot 2^{128}$ random outputs for there to be a 50% probability of a collision occurring, i.e. equivalent security to a 128-bit block cipher.